NYC Airbnb Market Analysis — Project Report

Uncovering pricing patterns and regulatory signals in New York City's short-term rental market

PYTHON · POSTGRESQL · POWER BI

01 — EXECUTIVE SUMMARY

Analyzed 20,770 NYC Airbnb listings across 22 attributes using Python, PostgreSQL, and Power BI. Cleaned significant data quality issues — including corrupted identifiers, missing values, and text-corrupted numeric fields — then uncovered a regulatory pattern tied to NYC's Local Law 18: 81.1% of listings default to a 30-night minimum stay, aligning closely with license data showing 84.6% of listings hold no formal short-term rental license. Delivered findings through a documented Python cleaning/EDA pipeline, a 10-query SQL analysis set, and an interactive Power BI dashboard.

02 — PROBLEM STATEMENT

New York City's Local Law 18 requires short-term rental hosts (stays under 30 nights) to register with the city. Hosts can avoid this requirement by setting a 30-night-or-longer minimum stay.

> **Core question:** Is this regulatory avoidance behavior visible in the data itself — and what does it, along with general pricing and availability patterns, reveal about the NYC Airbnb market?

03 — OBJECTIVES

- Clean and validate a real-world, imperfect dataset
- Characterize NYC Airbnb price distribution and its drivers
- Test whether license status relates to minimum-stay requirements
- Identify "ghost listing" activity patterns
- Deliver findings via Python, SQL, and an interactive dashboard

04 — DATASET OVERVIEW

20,770 listings across **22 columns**

Key fields: price, location (borough/neighbourhood), room type, minimum nights, license status, availability, reviews, rating, bedrooms, beds, baths

Market composition:

Manhattan 38.8% · Brooklyn 37.2% · Queens 18.1% · Bronx 4.6% · Staten Island 1.4%

Entire home/apt 55.6% · Private room 42.4% · Shared room 1.4% · Hotel room 0.5%

05 — METHODOLOGY

STAGE	TOOL	PURPOSE
Cleaning & EDA	Python (pandas, matplotlib, seaborn)	Inspect, clean, explore
Relational querying	PostgreSQL	SQL-based analysis
Dashboard	Power BI	Interactive stakeholder view

06 — DATA CLEANING: KEY ISSUES RESOLVED

Corrupted identifiers: 36.4% of `id` values damaged by Excel's scientific-notation conversion during a prior export — converted to text to prevent further precision loss and accidental numeric aggregation

Missing data: 34 rows missing price (including 7 nearly-empty rows) — checked for bias before dropping; missingness skewed toward Brooklyn (73.5% of missing-price rows vs. 37.2% baseline) but volume was small enough (0.16%) to drop safely

Duplicates: 12 true duplicate row pairs (24 rows) identified and removed — distinguished from ~8,000 rows sharing a corrupted `id` value, which were NOT duplicates

Text-corrupted numeric columns: `rating` (18.1% held placeholder text "No rating"/"New"), `bedrooms` ("Studio" mapped to 0), `baths` ("Not specified" → missing) — converted to usable numeric fields

07 — PRICE ANALYSIS

Median price: $125 · Mean price: $187.7 — the gap indicates strong right-skew

IQR: $80–$199 (middle 50% of listings)

6.7% of listings (1,383) statistically flagged as outliers above the $377.50 upper fence

8 listings priced at suspicious round-number extremes ($10,000–$100,000) — flagged as likely data errors, not genuine luxury pricing

Price by borough (median): Manhattan $150 · Brooklyn $125 · Queens $98 · Staten Island $99 · Bronx $89

Price-per-bed by borough (median, outliers excluded) — confirms the Manhattan premium holds even after controlling for listing size: Manhattan $109 · Brooklyn $80 · Queens $65 · Bronx $62.50 · Staten Island $60

08 — THE REGULATORY PATTERN: LOCAL LAW 18

81.1% of listings default to exactly a 30-night minimum stay — far beyond what organic guest demand would produce, consistent with hosts avoiding short-term rental registration

License status breakdown: No License 84.6% · Exempt 10.3% · Licensed 5.1%

Cross-referencing license status against the 30-night minimum shows the two patterns align, supporting the theory that the minimum-stay default functions as a registration workaround

09 — AVAILABILITY & LISTING ACTIVITY

`availability_365` shows a **bimodal distribution** — not a normal spread

11.9% of listings fully blocked (0 days available/year)

12.9% of listings fully open (365 days available/year)

This pattern suggests many listings function as binary on/off decisions rather than graduated availability — some may be inactive "ghost listings" still present on the platform

10 — REVIEW ACTIVITY

`number_of_reviews` (all-time): median 14 · mean 42.6 · max 1,865 (heavily right-skewed)

`number_of_reviews_ltm` (last 12 months): median 3 · mean 10.8

`reviews_per_month` (rate): median 0.65 · mean 1.26

Listings with very high review counts are almost exclusively lower-priced — expensive listings accumulate reviews more slowly due to lower booking frequency

11 — CORRELATION FINDINGS

Price shows **no meaningful linear correlation** with any numeric variable (all under 0.07) — price is driven by categorical factors (borough, room type), not continuous variables

`reviews_per_month` and `number_of_reviews_ltm` correlate strongly (0.85), as expected — both measure recent guest activity from different angles

Caveat: correlation only captures linear relationships — the 30-night clustering and bimodal availability are real, meaningful patterns that don't show up in this analysis since they're threshold effects, not straight-line trends

12 — FEATURE ENGINEERING

`price_per_bed` — normalizes price by listing size for fairer comparison

`price_flag` / `min_nights_flag` — flags extreme, likely-erroneous values without deleting rows

`is_30_night_min` — boolean flag testing the Local Law 18 hypothesis

`license_status` — bucketed license field (Licensed / Exempt / No License)

13 — SQL ANALYSIS

10 queries built against the cleaned PostgreSQL table, covering: aggregation, window functions, and CTEs — including a direct test of the license-status × 30-night-minimum hypothesis and a borough-vs-neighbourhood price comparison.

14 — POWER BI DASHBOARD

Interactive dashboard including:

- Average price by borough
- Listing count by borough
- Average monthly reviews by room type
- 30-night minimum by license status (testing the core hypothesis visually)
- Share of listings by borough
- Listing count by license status
- Total listings by number of bedrooms

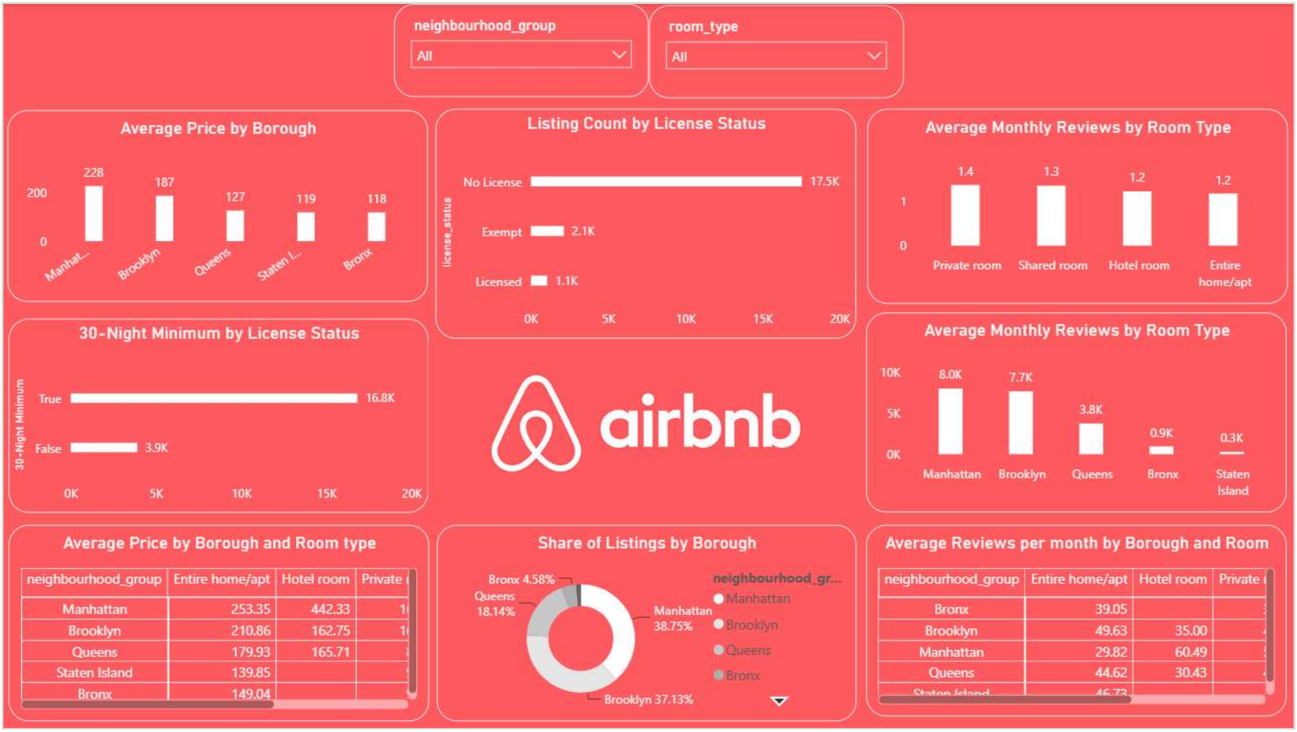

FIG. 1 — POWER BI DASHBOARD, FULL VIEW

15 — KEY FINDINGS SUMMARY

1. NYC Airbnb pricing is heavily right-skewed; Manhattan commands a premium that holds even after controlling for listing size
2. 81% of listings default to a 30-night minimum stay, aligning with an 84.6% "No License" rate — a visible fingerprint of Local Law 18 avoidance behavior
3. Availability is bimodal, not normally distributed, suggesting many listings are either fully active or effectively dormant
4. Price does not correlate linearly with review activity, availability, or host listing count — it's driven by location and room type, not booking behavior

16 — LIMITATIONS

- Dataset represents a single point-in-time snapshot, not longitudinal data — cannot show how the market changed before/after Local Law 18 enforcement
- Correlational, not causal — the license/minimum-stay relationship suggests but doesn't prove host intent to avoid regulation
- `id` corruption (36.4%) limits the ability to track individual listings reliably

17 — FUTURE WORK

- Convert `last_review` to a proper date type for time-based analysis
- Build a "listing activity" composite feature combining recent reviews and availability
- Extend SQL analysis with additional window-function-based ranking queries
- Add small-multiples visualization to Power BI for license status by borough